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A Mean Model Based Incremental Learning Technique for Extreme Learning Machine

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Abstract

Development of efficient methods for processing the online data is interesting as well as challenging in the computing scenario of data intensive applications, especially in streaming. Because of the inherent computational nature of Extreme Learning Machine, the non-iterative learning mechanism, it has undergone serious studies and exploitation especially in the domain of online data analysis. This paper gives a simple and effective strategy for updating the learned parameters, the knowledge of a machine learning system, in a computationally better perspective comparing with the existing state-of-the-art learning techniques. The performance of different measures for the central tendency is analyzed in this work. It is found that the results are comparable in respect of normal models. The overall performance of the proposed method with LIBSVM data sets gives an interesting direction for the improvement of learning machines for online data processing.

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1. Introduction

Real-time algorithms are always challenging because of the time constraints that have to be considered along with the normal computational aspects. The volume of data that has been generated in every second is crossing the limits

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of online computing facilities. Because of the huge volume of data accumulated in the digital world, the processing of data on its arrival has that much importance at present scenario. The developments in online data management have increased the reliability of lots of applications in the different areas in our daily life such as video analysis and surveillance [1,2], medical and health care[3,5], network and security[4,6,7]. The working pattern of the online algorithms mainly focuses on the processing of data either in sequence or in batch-wise. In both cases, the quantity of data will increase periodically, and the entire data will not available for processing. The execution of normal offline methods for extracting the knowledge is expensive and impractical for online data processing. The intense researches in this area lead to the online extensions of various machine learning methods, but all of them could not retain the performances [8]. The incremental learning methods in which the knowledge derived from a batch or sequence of data may be updated with the features of another segment of data in a periodic manner. Incremental learning algorithms have a vital role in the processing of a large quantity of online data [9,10].

Joao et al.[11] put forwarded a new ensemble based algorithm for classification called Iterative Boosting Streaming ensemble (IBS). In IBS, the model is updated periodically and according to the arrival of concepts that have significant changes with the existing one. The proposed model is made lenient by providing the mechanism to change the change the basic learner as per the performance of the model for that particular instant. It is found that the updating rate in IBS is considerably very low while dealing the stream with similar concepts. Yanxia et al.[12] gives an online bagging ensemble method for stream data learning. In this classifier, author efficiently presented online re-sampling technique on the input data with the help of a coding method. Through this approach, they could minimise the impact of correlations between the classifiers and unnecessary updating of model.

In many real-time and online applications, normal learning algorithms are not preferred because of the iterative mode of learning. Comparing with other learning methods, the working principle of ELM is simple, and the meantime it is computationally efficient [13]. The extensive growth of this single layer, non-iterative learning method shows its acceptance in the machine learning researches. The twin extreme learning machines [14] and its variants used in the pattern classification [15], multilayer sparse ELM used for processing hyperspectral images [16] are some of the latest developments in ELM. The deep weighted ELM [17], motivated from the deep network, shows the interventions of researchers to enhance the ELM based learning algorithms.

An incremental learning model for the Extreme Learning Machine (ELM) is proposed in this work. Instead of giving random values as initial weights, SVD hidden nodes are used to prepare the ELM model [18]. The learning parameters obtained in each processing are collectively used to prepare the new candidate sets for the entire dataset. The experiments with different candidate parameter, final weights, selection procedures show that potential of advanced researches in this specific area.

The rest of the paper is structured as section two is dedicated to explaining the basic theory and developments in the ELM based learning methods. The proposal of this paper is explained in section three. The experiments and the results obtained are given in the following section. The analysis of the results is also given in the fourth section. The overall conclusion of the work is given in the final section.

2. Theory and practices of extreme learning machine

Distinct from the typical concepts of machine learning algorithms, the Extreme Learning Machine [19] is a single-layered noniterative feed-forward neural network. As a result of the new paradigm, the ELM could do better in terms of resource utilized and its execution time.

Consider a dataset that consists of $X_{NK} = \{x_1, x_2, x_3 \dots x_N\}$ and $Y_{N1} = \{y_1, y_2, y_3 \dots y_N\}$, where N is the number of instances in the dataset, K is the total number of parameters considered in each x_i and Y is the target vector of X. The training in ELM is done with a set of the randomly selected weight vector. The process of optimizing the initial weight by minimizing the error to produce the expected outcome is treated as learning.

$$\min_w \|HW - Y\|_F^2 \quad (1)$$

Where W is the output weight after completing the learning activity. H is the output of the hidden layer.

$$H_{NL} = \begin{bmatrix} g(w_1, b_1, x_1) & \dots & g(w_L, b_L, x_1) \\ \vdots & \ddots & \vdots \\ g(w_1, b_1, x_N) & \dots & g(w_L, b_L, x_N) \end{bmatrix} \quad (2)$$

L is the number of hidden nodes, $g(\cdot)$ is the activation function, w_i, b_i are the input weight and bias of i^{th} node in the model.

The output weight W is derived as

$$W = H^+ Y \quad (3)$$

where H^+ is the Moore–Penrose generalized inverse of H .

There are different variations in ELM, and the scientists are trying to tune it as an alternative method for machine learning algorithms

The kernel based ELM models, especially the Multi-Kernel ELM [20], are designed for tuning the weights as well as other parameters for the model.

Twin ELM used the concepts of Generalized Eigenvalues Proximal SVM [14] where the data set is partitioned into two separate sets and weights are derived for both. The separating hyperplanes are created for the input data x as

$$f_1(x) = \beta_1 \cdot h(x) = 0 \quad (4)$$

$$f_2(x) = \beta_2 \cdot h(x) = 0 \quad (5)$$

The class of the data set x is determined from the distance between the hyperplane and data point x ,

$$d(x) = \arg \min_{r=1,2} d_r(x) = \arg \min_{r=1,2} |\beta_r^T h(x)| \quad (6)$$

There is a bunch of variations of ELM, specifically designed for online algorithms, which required regular updating of the existing knowledge. The incremental learning algorithm put forward by Huang et. Al [21] is one of the earliest and proven methods in this category. There are lots of scientists who worked on the top of this method and come up with certain variations as EI-ELM [22] and CI-ELM[23]. The mean time, separate works were carried out for real-time algorithms and the major proposal came from Liang et al [24]. Obviously, the variants EOS-ELM [25], OS-KELM [26] and OS-FELM [27] came from Liang et al only.

Another strategy for dealing the online data management is batch wise processing, and there are couple of successful attempts as well. The Ensemble ELM [28] is the pioneer in this strategy in which the input data set is divided into B batches. In each iteration, $(B-1)$ batches are used for learning and other batch used for testing. The parameters obtained in each iteration are updated with mean and norms. Apart from the results, they could lead a major role in the ensemble based methods in ELM through this work. The Optimally Pruned ELM(OP-ELM)[29] and Ensemble based Reactivated Regularization Extreme Learning Machine(ER2-ELM) [30] are some among them.

3. Proposed Work

The proposed method in this paper is designed for online data processing and it is motivated by the Ensemble based ELM [28]. As explained in the previous sections, the data set X with N instances and K number of parameters is used for experiments. The data set is also having a target vector of N size. y_i represents the class in which the i^{th} instance belongs, $y_i \in \{c_1, c_2, c_3 \dots c_p\}$, since there are p classes in the dataset. The data set is further partitioned into 80:20 ratios, if a separate testing set is not available, for training and testing purposes.

The input dataset which has portioned for training, S_{RK} , is further segmented into l blocks.

$$S_{RK} = \{b_1, b_2, b_3 \dots b_l\} \quad (7)$$

As suggested by Yihe Wan et. Al [18], the singular value decomposition of b_i is calculated. The initial weight and bias values are taken from the SVD decomposed matrices.

Using the (2) and (3), the output weight w_i is derived and stored in the reservoir R for further reference.

$$R = \{w_1, w_2, w_3 \dots w_l\} \quad (8)$$

$$f(R) = \beta \quad (9)$$

β is the net output weight of the entire collection R and $f(\cdot)$ represents the centrality measure of the collection.

Because of the inherent features of the ELM model, the time of execution and resources used for the computation are very less and promising for online data processing. In the proposal, the learning set is further divided into smaller blocks; the execution time will be negligible in comparison with the process for the entire dataset.

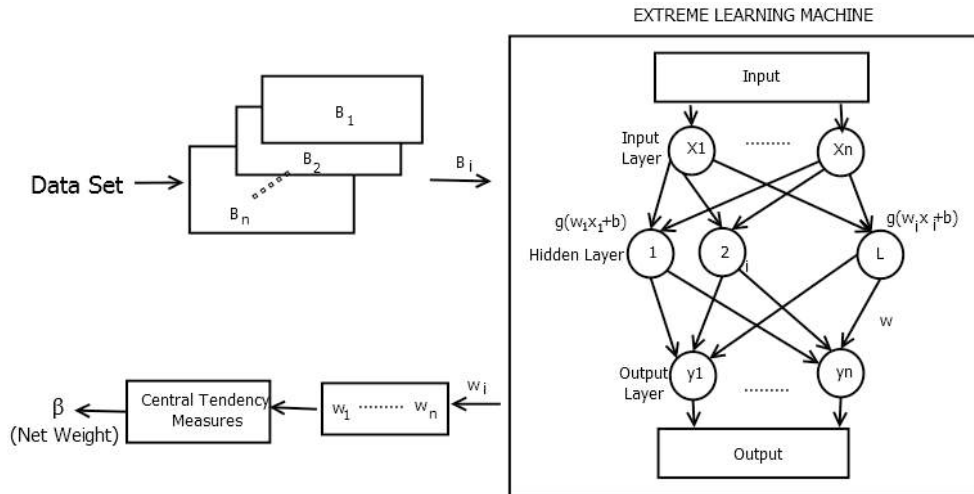


Fig. 1. The working of the proposed method for the dynamic updating of net weight

The Fig. 1 shows the overall working of the proposed method. The training set from the data set is initially divided into n batches. At an instance i , the batch B_i will be given for the learning activity and the output derived by the ELM is used to update the existing learning parameters.

4. Experiment and Results

A set of experiments were done to check the reliability and accuracy of the proposed method. Two datasets are used in the entire experiments- Real-Sim and RCV1[31]. These datasets are available in the LIBSVM repository [32] for machine learning algorithms. There are 72,309 instances in the realism and 697,641 instances in the RCV1. There are 20,958 and 47,236 parameters in Real-Sim and RCV1 datasets respectively. Separate datasets for testing is available in RCV1 and in the case of Real-Sim, it is divided for testing and training purposes.

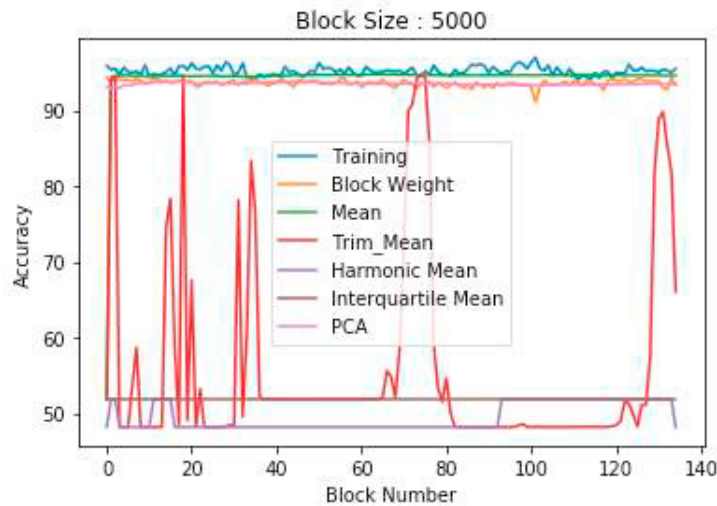


Fig. 2. Performance of the different centrality measures with RCV1 dataset.

As explained in the previous section, the individual output weight w_i is stored in R . Our initial experiment was to identify the centrality measures that can be used for deriving the net output weight β . The common centrality measures such as Mean, Trimmed Mean, Harmonic Mean, Interquartile Mean are measured in each iteration. The result obtained in the experiment for the dataset RCV1 is shown in the Fig.2

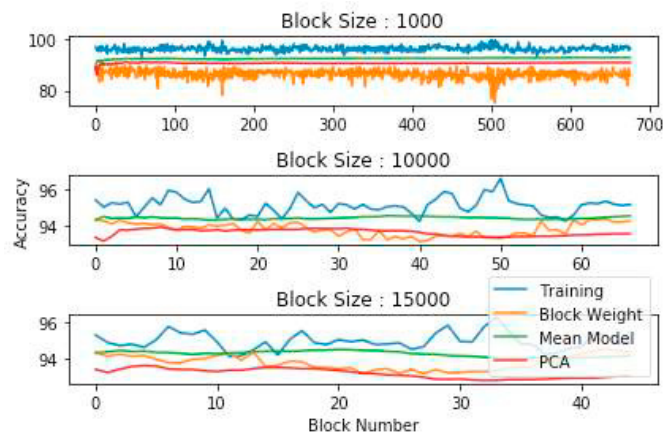


Fig. 3. Results of the experiments with RCV1 data sets

It is noted in the experiment that the measures Trimmed mean, harmonic mean and Interquartile Mean provides inconsistent results throughout the experiments. Hence those measures are excluded in the later exercises. The performance of PCA, and mean model are found appealing. The weight obtained for each iteration, the block weight, is also used for the testing purposes.

The results of the other experiments given in Fig. 3 and Fig.4 tell that the mean model for updating the net output weight gives an appealing direction. The principal component of the collection also gives comparatively better results due to the underlined theory of PCA.

The Fig.4 depicts the performance of the mean model and PCA in the weight updating process. In this experiment, the RCV1 data set is used. The data set is divided into blocks of different sizes for the experiments in order to simulate

the nature of online arrival of data. It is clear that the training accuracy of the ELM model shows some vibrant fluctuations.

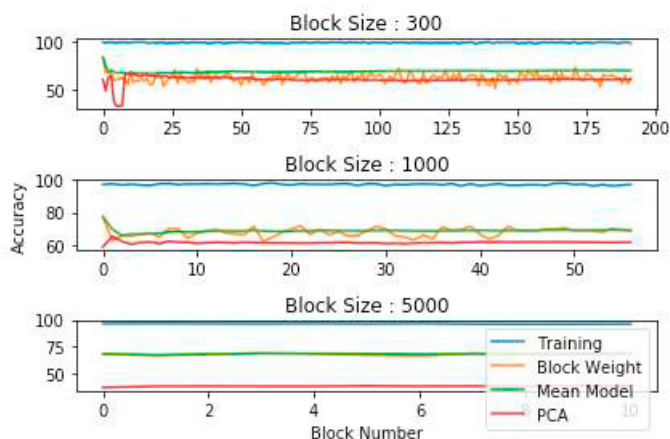


Fig. 4. Results of the experiments with Real-Sim data sets.

It is clear in the experiments that the mean model of centrality measure records more consistency. It gives almost the same training accuracy in some places along the graph path. The accuracy achieved by the mean model is always recorded above the batch weight and principal component.

The experiment with Real-Sim data set also shows the same pattern of the results. It is clear in the Fig.5 that as in the case of the previous experiment, the position of the mean model is above the block-wise weight. The size of the RCV1 data collection is very large compared with Real-Sim. The accuracy of the mean model in this experiment is found less than the previous experiment, but the consistency of the result shows the reliability of the model.

5. Conclusion

The capabilities of Extreme Learning Machine in online data processing are explored in this paper. The features of ELM, single layer non-iterative learning model, are experimentally proven and suitable for real-time applications. In the case of online applications, where the data should be processed before it is stored, the learning parameters that have been generated for an instance should be updated without causing much computational burden. The proposal in this paper suggests that the mean model of centrality measure is a good method to update the net weight. It is found in the experiments with a couple of publically available datasets that the mean model is always maintained an upper hand in the accuracy of prediction at any instance of the experiment. The consistency of the mean model shows its reliability. The future works in this direction can give more improvements in the online data processing algorithms.

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